

Recurrent Disease in Kidney Transplantation

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Kidney transplantation is a treatment, not a cure. Although transplantation can restore kidney function, it rarely removes the cause of the original kidney disease and recurrence is dreaded by patients and clinicians alike. Globally, glomerulonephritis (GN) and diabetes are the two leading causes of kidney failure and are the most common primary diseases in patients undergoing kidney transplantation. That both GN and diabetes can recur after transplantation is a source of consternation for recipients and clinicians.

Recurrence is typically a time-dependent phenomenon. The likelihood of recurrence increases as the graft ages. With increased graft survival rates over the past 30 years, the apparent incidence of recurrence has grown.^{1,2} An analysis of U.S. Renal Allograft Disease Registry data examining GN recurrence demonstrated a prevalence of 2.8% at 2 years, 9.8% at 5 years, and 18.5% at 8 years of follow-up after transplantation, and such patients were twice as likely to experience graft failure compared with those without recurrence.³

Recurrence has a powerful adverse impact on transplant survival, which is increasingly apparent as time since transplantation lengthens. In an analysis of data from the Australia and New Zealand Dialysis and Transplant Registry (ANZDATA) including more than 7968 patients with GN who received a kidney transplant, biopsy-proven recurrence was found in approximately 10% over the median follow-up time of 8.6 years, with 45% of those losing their graft within 5 years of the recurrence² (Fig. 113.1). For those with GN, recurrent disease is the third most common cause of graft loss after the first year after transplantation, behind death with a functioning graft and chronic rejection (CR) but substantially ahead of acute rejection (see Fig. 113.1).⁴ However, for those with known recurrent disease, it is the most common cause of graft loss.²

De novo GN and new-onset diabetes after transplantation (NODAT) may affect the transplanted kidney, although both are relatively uncommon causes of graft failure and may be difficult to distinguish from recurrence or from CR. Like recurrent disease, the prevalence of both appears to increase with time after transplantation. Given the high incidence of NODAT and increasing graft survival in recent eras, de novo diabetic nephropathy may become a significant clinical problem in the future. However, the major impact of NODAT in the first 10 years after transplantation is an increase in cardiovascular (CV) mortality with little impact on death-censored graft failure.⁵

DEFINITIONS

Diagnosis of recurrence requires histologic demonstration of the same disease involving both the native and transplanted kidneys. Diagnosis of recurrence causing graft failure requires a clinical decision that recurrence was the dominant contributor to graft loss (other contributors, such as CR, may be present).

The incidence of recurrent disease is probably underestimated because a histologic diagnosis of the primary kidney disease is not always obtained, and many transplant biopsies are not done with recurrent disease in mind, so immunohistologic and electron microscopic (EM) examination (mandatory for optimal diagnosis of recurrent disease) may not be undertaken.¹ In addition, a clinical diagnosis of CR is often made in patients with declining graft function and proteinuria, so that recurrent disease is not properly excluded.

Additional factors confound the available evidence. Many reports of disease recurrence are retrospective, single-center studies. Recall bias and incomplete documentation, changes in practice over time, and peculiarities of local patient populations and local practices may limit relevance to other populations. The most definitive reports have come from analyses of the large registry databases of Europe, the United States, and Australasia (the latter registry being known as ANZDATA). Registries capture data on large numbers of patients but are still subject to bias because of factors including unit participation rates, quantity and type of data collected, accuracy, consistency of reporting, and reliability of data entry. How recurrence is defined and diagnosed and which outcomes are measured is crucial. For example, immunoglobulin A nephropathy (IgAN) recurred in 58% of patients in one series in which all recipients underwent biopsy,⁶ but in approximately 25% of cases in which biopsy was performed only when clinically indicated.⁷ When graft loss from IgAN recurrence is the outcome measure, the risk decreased to approximately 10% at 10 years of follow-up.⁴ Thus, definition of recurrence, outcome measures, study design, era, and source of data all need to be considered in assessing the published literature.

RECURRENT GLOMERULONEPHRITIS

Many types of GN may recur after transplantation; however, the rate and consequences of recurrence vary enormously. For example, anti-glomerular basement membrane disease (anti-GBM; Goodpasture disease) recurs only rarely, but when it does, it is likely to cause rapid graft loss. In contrast, C3 glomerulopathy (formerly known as dense deposit disease and/or type II membranoproliferative glomerulonephritis [MPGN]) recurs in over 80% of patients; however, the disease tends to be very slowly progressive, and graft survival beyond 10 years is typical. Recurrence of focal segmental glomerulosclerosis (FSGS), IgAN, and membranous nephropathy (MN) are the most frequently encountered clinical problems (Fig. 113.2).

Several factors may influence the risk for recurrence in addition to the type of GN. Time since transplantation is clearly important and may be related to the duration of graft exposure to the nephritogenic factors responsible for GN.¹ Those grafts that survive long term are exposed to nephritogenic factors for longer and are more likely to develop recurrent GN. Consistent with this, recipients of human leukocyte antigen (HLA)-identical transplants rarely experience rejection

Recurrent Glomerulonephritis and Other Causes of Kidney Allograft Failure

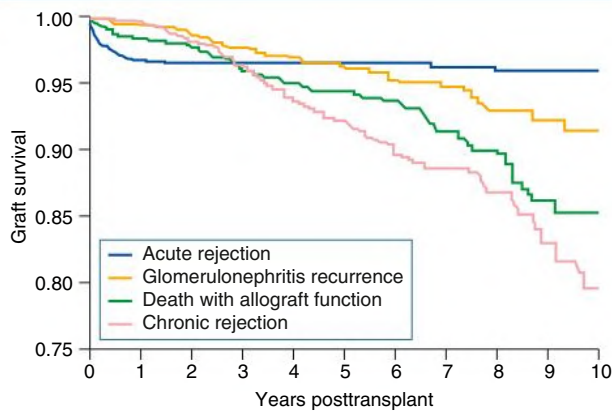


Fig. 113.1 Recurrent Glomerulonephritis (GN) and Other Causes of Kidney Allograft Failure. Kaplan-Meier analysis of the relative contributions of acute rejection, GN recurrence, death, and chronic rejection to graft loss during the first 10 years after transplantation among patients who underwent transplantation because of end-stage kidney disease caused by GN. (Modified from Briganti EM, Russ GR, McNeil JJ, Atkins RC, Chadban SJ. Risk of renal allograft loss from recurrent glomerulonephritis. *N Engl J Med*. 2002;347[2]:103–109.)

Primary Glomerulonephritis and Kidney Allograft Failure

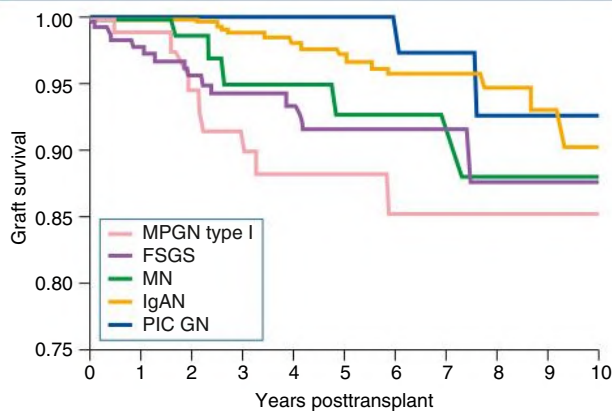


Fig. 113.2 Primary Glomerulonephritis (GN) and Kidney Allograft Failure. Kaplan-Meier analysis of freedom from graft loss caused by recurrent GN during the first 10 years after kidney transplantation among patients with a primary diagnosis of GN. FSGS, Focal segmental glomerulosclerosis; IgAN, IgA nephropathy; MN, membranous nephropathy; MPGN, membranoproliferative glomerulonephritis; PIC GN, pauci-immune crescentic glomerulonephritis. (Modified from Briganti EM, Russ GR, McNeil JJ, Atkins RC, Chadban SJ. Risk of renal allograft loss from recurrent glomerulonephritis. *N Engl J Med*. 2002;347[2]:103–109.)

and enjoy prolonged graft survival but have a high rate of recurrent GN.⁸ In one report of HLA-identical recipients, recurrent GN was present in 36% to 42% of those in whom biopsy was performed and resulted in 24% of graft losses, being the second most frequent cause of graft loss after death in this group.⁸ Patients sustaining first graft loss from recurrent GN are also at higher risk for recurrence in a subsequent graft.

Improvements in immunosuppression over the past 40 years have led to reduced rates of acute rejection. Such improvements also may have had an impact on rates of graft loss from IgAN recurrence, which

has declined by nearly 50% over the past 10 years according to one ANZDATA registry analysis.⁹

U.S. Renal Data System (USRDS) registry data failed to demonstrate that any individual immunosuppressive agent is superior for preventing recurrence.¹⁰ Nevertheless, the immunosuppressive regimen may be important. For example, one small retrospective study found that induction therapy with rabbit antithymocyte globulin was associated with a lower risk for IgAN recurrence compared with either anti-CD25 antibodies or no induction.¹¹ Corticosteroid withdrawal may lead to increased rates of recurrence,¹² especially in recipients with IgAN, in whom a twofold increase in the risk for graft failure from recurrence of IgAN has been reported.⁹ Similarly, inclusion or addition of cyclophosphamide to immunosuppressive regimens may treat recurrent vasculitis affecting the graft.¹³ Sirolimus or everolimus may result in proteinuria and kidney dysfunction, especially in patients with underlying GN, suggesting that mammalian target of rapamycin (mTOR) inhibition may accelerate preexisting glomerular injury in the graft.¹⁴ Effects of individual agents are likely to be disease-specific, but the necessary data remain sparse.

Strategies to reduce the risk for recurrence have been reported. Bilateral native nephrectomy to eliminate persistent antigenic stimulation appears unhelpful; indeed, nephrectomized patients experienced a higher incidence of recurrence compared with those with native kidneys left in situ in one large single-center retrospective study.¹⁵ Induction of disease remission before transplantation and prolonged time on dialysis pretransplantation (both aimed at permitting disease “burnout”) do not appear to be effective except in the case of anti-GBM disease, in which a delay in transplantation until the patient has been serologically negative for at least 6 months virtually eliminates the risk for recurrence.⁴ Avoiding living related donation to recipients with FSGS has been debated for years but overall appears not to have any impact on risk of recurrence.¹⁶

As with GN in native kidneys, proteinuria, deterioration in kidney function and, to a lesser extent, hematuria are the cardinal manifestations of recurrent GN.¹⁷ The pattern of kidney and extrarenal manifestations is frequently similar to that of the native disease, except that the overall rate of progression may be slower. Extrarenal features of the primary condition may recur, such as thrombocytopenia and hemolysis in hemolytic uremic syndrome (HUS) and extrarenal vasculitis in recurrent antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis. Serology may be helpful in some patients, such as anti-GBM antibody detection in those with Goodpasture disease, but not necessarily in others, such as those with recurrent lupus nephritis (LN).

The differential diagnosis of recurrent GN is clinically important because it may influence management (Table 113.1). Chronic rejection and diabetic nephropathy, recurrent or de novo, may manifest with progressive graft dysfunction, proteinuria, and hypertension and may therefore be clinically indistinguishable from recurrence. De novo GN also should be considered. Viral diseases of the kidney should be considered (particularly BK nephropathy) because reducing immunosuppression, and in selected instances, antiviral drugs may provide benefit. Obstructive uropathy and tumors involving the graft should be excluded by an ultrasound scan. Finally, recurrence may coexist with chronic rejection or calcineurin inhibitor (CNI) toxicity. Indeed, every condition that can lead to chronic graft dysfunction should be considered in the differential diagnosis of recurrence (see Table 113.1 and Chapters 109 and 110).

Histologic evidence of recurrence is required in all patients. Biopsy can provide the diagnosis, exclude alternative diagnoses that may require different approaches to treatment, and provide important prognostic information pertinent to the affected graft and also relevant to any future consideration of retransplantation. Full evaluation of

TABLE 113.1 Differential Diagnosis of Recurrent Glomerulonephritis

Diagnosis	Frequency and Timing	Clinical Features	Laboratory Features	Biopsy Features	Management
Recurrent glomerulonephritis	Common; variable timing, days to years	Proteinuria, hematuria, kidney impairment, hypertension	Similar to primary glomerulonephritis; serology may be negative	Same as primary glomerulonephritis ⁵	Disease specific
De novo glomerulonephritis	Uncommon; variable timing but typically later than recurrence	Proteinuria, hematuria, kidney impairment	Type specific	Type specific ^{5,8}	Antiproliferation strategies (see Chapter 82)
Chronic rejection	Very common; increasing incidence with time	Hypertension, proteinuria, kidney impairment Calcineurin inhibitor exposure		Tubulointerstitial fibrosis, arteriolar hyalinosis, transplant glomerulopathy	Minimize calcineurin inhibitor and antiproliferation strategies (see Chapter 82)
Graft pyelonephritis	Uncommon; typically early after transplantation	Fever, pyuria, kidney impairment	Positive blood or urine cultures	Neutrophil infiltration	Antibiotics
BK nephropathy	Uncommon; typically 1–5 y after transplantation	Kidney impairment, decoy cells in urine	Serum BK PCR positive	Tubulitis with tubular cell atypia and inclusions, normal glomeruli	Minimize immunosuppression and consider antiviral drugs
Acute rejection	Common; early	Kidney impairment, oliguria	Nonspecific	Tubulitis with or without vasculitis ¹⁵	Increase immunosuppression
Kidney tumor/PTLD	Uncommon, rare; early or late	Kidney impairment, kidney mass	Anemia, EBV positive	Atypical cells, mitoses, monoclonality	Minimize immunosuppression, consider chemotherapy

EBV, Epstein-Barr virus; PCR, polymerase chain reaction; PTLD, posttransplantation lymphoproliferative disease.

the biopsy specimen by light microscopy, immunohistology, and EM is desirable and, in many patients, essential to confirm recurrence.¹⁸ Light microscopy and immunohistology are necessary to differentiate recurrent from de novo GN, rejection, and CNI toxicity. The presence of tubulitis should suggest acute rejection. Chronic rejection may produce chronic interstitial inflammation and transplant glomerulopathy, which may be indistinguishable from MPGN on light microscopy (Fig. 113.3; see also Chapter 109). The use of immunohistology to define the immunoglobulin and complement component content of immune deposits and EM to establish the structure of basement membrane and location of deposits may clarify the diagnosis.¹⁸ The risk of recurrence for common histologic types of kidney disease is summarized in Table 113.2.

RECURRENCE OF SPECIFIC GLOMERULAR DISEASES

Immunoglobulin A Nephropathy and IgA Vasculitis (Henoch-Schönlein Purpura)

IgAN is the most common form of GN leading to kidney failure, and affected patients frequently become transplant recipients. Histologic recurrence is frequent and increases with time since transplantation. Published recurrence rates vary greatly (from 8%–53%). Much of this variation appears to be because of study differences in biopsy indication (protocol or clinical) and length of posttransplant follow-up.¹⁹ The highest published recurrence rates are from centers performing protocol biopsies in addition to those that are clinically indicated. This is likely because the early stages of IgAN recurrence are frequently not accompanied by clinical changes, such as proteinuria, hematuria, or graft dysfunction.¹⁹

Recurrence is difficult to predict. No large prospective multicenter cohort studies demonstrate risk factors for histologic and/or clinical recurrence. Single-center studies and registry analyses suggest that younger age at transplantation, rapid progression of the native IgAN, degree of proteinuria, and donor factors (including HLA matching) may be associated with higher risk.¹⁷ The suspicion that living donor grafts have higher rates of recurrence than deceased donors has not been confirmed.¹⁷ Thus, patients with IgAN should not be precluded from consideration for living donor transplantation. The number of HLA mismatches between donor and recipient also may have a role in recurrence rates. Two Australian registry studies have found that those with zero mismatch kidneys have higher rates of recurrence than those with one or more HLA mismatches.^{9,20} Similarly, ABO-incompatible transplants have been found to have lower rates of recurrence, possibly because of differences in immunosuppression regimens.²¹

Choice of immunosuppression after transplantation is controversial. One large USRDS registry analysis suggested that individual drug choices did not affect the risk for graft loss attributed to recurrence.¹⁰ In contrast, a retrospective analysis of ANZDATA found that continuation of corticosteroids was strongly associated with protection from graft loss caused by recurrence of IgAN, although not from other types of GN.⁹ Observational data also suggest that induction with antithymocyte globulin may afford relative protection.¹¹

The clinical expression of recurrent disease is variable and time dependent. Graft loss within the first 3 years after transplantation is uncommon (see Fig. 113.2), although it can occur, particularly when IgAN in the native kidney was rapidly progressive or after previous graft loss from recurrence.⁷ An Italian cohort study of 190 patients followed for 15 years reported that of the 22% of patients who developed recurrence, almost a third ultimately lost their graft because of IgAN

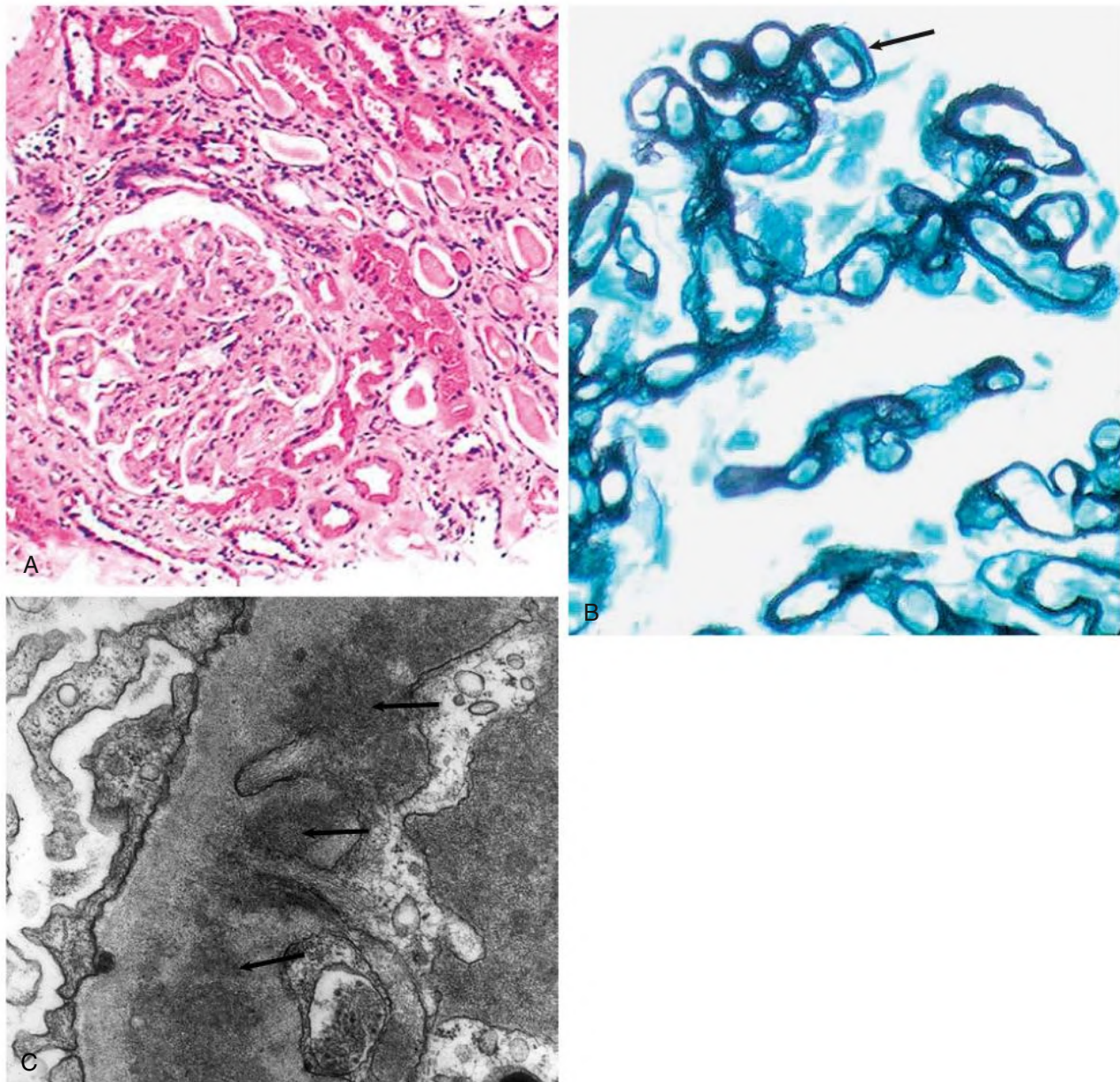


Fig. 113.3 Transplant Glomerulopathy and Membranoproliferative Glomerulonephritis (MPGN). Transplant biopsy specimen from a patient with end-stage kidney disease caused by biopsy-proven idiopathic MPGN type I, who received a kidney transplant and had a progressive reduction in glomerular filtration rate with proteinuria 1.5 g/day and hypertension. (A) Light microscopy. Glomerular hypercellularity and lobulation on a background of chronic interstitial inflammation and fibrosis, with protein casts within dilated tubules. (Hematoxylin-eosin stain; $\times 100$.) (B) Subendothelial deposits and basement membrane reduplication (*arrow*). (Methenamine silver stain; $\times 400$.) (C) Electron microscopy showing subendothelial electron-dense deposits (*arrows*) ($\times 7500$). There were also prominent C3 deposits on immunofluorescence (not shown). Light microscopy was therefore suggestive of recurrent MPGN but was also consistent with chronic rejection (CR) with transplant glomerulopathy associated with CR. Immunofluorescence and electron microscopy (subendothelial deposits) confirmed recurrence of MPGN. Compare these images with Figs. 112.3 and 112.4. (A–B, Courtesy Dr. Paul McKenzie, Royal Prince Alfred Hospital, Sydney, Australia.)

recurrence during follow-up.²² Death-censored graft survival at 15 years was approximately 10% lower in patients who had IgAN than in controls, largely attributable to IgAN recurrence.

Recurrence of IgA vasculitis (IgAV) is less well characterized but appears to be similar to IgAN. An analysis of the United Network for Organ Sharing (UNOS) database including 339 patients with kidney failure caused by IgAV who received a first kidney allograft reported a frequency of graft loss caused by recurrence of 13.6% and no difference

in 10-year graft survival between recipients with IgAN (58.4%) or IgAV (59.3%) as their primary kidney disease.²³

Treatment of recurrent IgAN and IgAV has not been systematically evaluated. The addition of corticosteroid maintenance in corticosteroid-free patients with recurrence is unproven but reasonable. A change to mycophenolate mofetil (MMF) or the use of fish oil, antiplatelet agents, and tonsillectomy cannot be recommended. The use of nonspecific measures to prolong kidney survival is appropriate,

TABLE 113.2 Recurrent Diseases in Kidney Transplants and Effects on Graft Survival

Disease	Clinical Recurrence Rate (%)	Graft Loss in Recurrent Disease (%)
Primary focal segmental glomerulosclerosis	20–50 (children), 10–15 (adults)	40–50
Membranoproliferative glomerulonephritis type 1	20–30	30–40
Dense deposit disease	80	20, often late
Hemolytic uremic syndrome (HUS)		
Classic D+ HUS	0–13	Uncommon
Atypical D– HUS	30–50	55–100
Familial HUS	57	Approaching 100
IgA nephropathy	30–40, increases with longer duration of follow-up (30%–60% histologic recurrence rate)	16–33
IgAV (Henoch-Schönlein purpura)	Rare (despite 50% histologic recurrence rate)	Rare
Membranous nephropathy	10–29 (histologic recurrence may be more common)	Up to 50
Systemic vasculitis, including granulomatosis and microscopic polyangiitis	10–20	20–50
Anti-GBM disease (Goodpasture disease)	<5	50
Systemic lupus erythematosus	1–30	Rare
Amyloidosis	25	10–20

GBM, Glomerular basement membrane; IgA, immunoglobulin A; IgAV, IgA vasculitis.

including tight blood pressure control, renin-angiotensin system (RAS) blockade, and avoidance of nephrotoxins.

Membranous Nephropathy

Histologic recurrence of MN can be found in up to 40% of grafts,²⁴ and for those with documented recurrence, graft loss rates of over 50% at 3 years of follow-up have been reported.^{25,26} Patients who have previously lost a graft to recurrence are at higher risk on retransplantation, but disease course, duration of dialysis, HLA genotype, graft source, and immunosuppression have not been found to predict recurrence risk.²⁶

In 2009 M-type phospholipase A₂ receptor (PLA₂R) was identified as the antigen targeted in 70% to 80% of idiopathic MN.²⁷ This has spurred interest in the use of anti-PLA₂R antibody titers in monitoring disease activity and predicting recurrence. A number of small single-center and multicenter observational studies suggested a role for the adoption of PLA₂R measurement as a clinical tool in transplantation, with higher levels at transplantation associated with higher rates of recurrence.^{28–31} The largest of these studies reported a recurrence rate of 57% among patients with primary MN, with a median time to recurrence of 4.1 months.²⁸ The positive predictive value of pretransplant anti-PLA₂R antibodies was 83% (10 of 12), and the negative predictive value was 42% (5 of 12).²⁸ Current data show that the presence of anti-PLA₂R at the time of transplant increases the risk for recurrence; however, both sensitivity and specificity are imperfect and larger studies are required. At present, there is insufficient evidence to recommend delaying transplantation until anti-PLA₂R antibody status is negative; however, rising titers should warrant caution and consideration of a delay in transplantation.

Management of recurrent MN is based on anecdotal reports and extrapolation of data on the management of native kidney MN. There has been some success with rituximab reported in small case series, but the findings await confirmation in a randomized trial. Spontaneous remission appears to be less common than is seen in MN affecting native kidneys. The cumulative exposure to immunosuppressive therapy should be considered because these patients may be at increased risk for lymphoma. Living donor transplantation appears warranted for first grafts but in our opinion should probably be avoided for second grafts if the first was lost early because of recurrence.

To diagnose recurrence requires a native kidney biopsy showing MN and a graft biopsy showing MN. This is especially important in MN, in which de novo MN has been reported in 2% to 15% of transplant recipients and tends to manifest more insidiously and later than recurrent MN.³²

Focal Segmental Glomerulosclerosis

FSGS recurs in 20% to 30% of first transplants.^{33–35} FSGS is a heterogeneous group of conditions and those with familial or sporadic forms associated with mutation of slit-diaphragm proteins such as podocin do not recur, although rare cases of posttransplant nephrotic syndrome caused by the development of antibodies directed against the “neoantigen” within the donor kidney have been reported. FSGS secondary to obesity, vascular disease, or reflux nephropathy and those with a very slow rate of progression are at very low risk for recurrence. In contrast, patients with primary FSGS and in particular those with an aggressive initial course (heavy proteinuria and kidney failure within 3 years of onset), age younger than 15 years, mesangial hypercellularity on biopsy, or with recurrence in a previous graft are at greatest risk.^{34,35} The rate of recurrence is higher than 75% in subsequent grafts when the first graft was lost because of recurrence.³⁶ Living related donor transplantation has previously been implicated as a risk factor for recurrence; however, a large USRDS analysis refutes this notion,¹⁶ and ANZDATA data show improved graft survival from living donors.³⁷

Recurrence may occur early (typically within the first month post-transplantation) and initially manifests with heavy proteinuria, followed by hypertension and graft dysfunction. Patients with recurrent disease appear more susceptible to acute rejection and acute kidney injury³⁸ and graft loss. Recurrence has been associated with early graft loss in up to 50% of patients^{33,34,37}; however, treatment with plasma exchange appears to have delayed graft loss in many patients and decreased the incidence of overall graft failure^{4,33} (see Fig. 113.2).

Primary FSGS appears to be caused by a circulating factor that targets podocytes, although the specific factor(s) involved remain unknown. Soluble urokinase plasminogen activator receptor (suPAR) and the costimulatory protein B7-1 (CD80) have been proposed to mediate recurrence; however, reports have been anecdotal and subsequent studies have failed to prove the role of either molecule or consistently demonstrate success with therapies targeting them.^{39–43}

Plasma exchange or immunoadsorption with either a protein A or anti-IgG column is effective for some patients who develop recurrent FSGS, presumably because it removes the circulating permeability factor.^{44,45} Although not studied in a randomized controlled trial (RCT), several series have reported disease remission in the majority of patients who receive treatment within 2 weeks of recurrence.^{45,46} The potential for positive publication bias and the absence of RCT evidence make prognostic information speculative; however, response rates appear likely to exceed 50% when therapy is commenced within 2 weeks of clinical recurrence, with lower response rates anticipated when therapy is delayed.⁴⁶ Given that graft survival appears to be significantly prolonged in patients who respond to therapy, a course of plasma exchange is warranted in all patients without contraindications (see Chapter 104). Immunosuppression should include a CNI, and the addition of an angiotensin-converting enzyme (ACE) inhibitor should be considered. A minority of patients with an incomplete response or relapse after cessation of initial therapy will require repeated or long-term plasma exchange⁴⁷ or concurrent treatment with secondary agents such as rituximab or cyclophosphamide.^{45,47–49} For such patients, we recommend a course of rituximab 500 mg weekly for 4 weeks, in addition to existing immunosuppression and weekly plasma

exchange. Pretransplantation plasma exchange has not been shown to be beneficial for preventing recurrence. See Fig. 113.4 for an approach to the management of recurrent FSGS.

Minimal change disease is a far less frequent cause of kidney failure. Recurrence of disease after transplantation has been reported; however, it is difficult to be sure that the underlying disease was not FSGS.

Membranoproliferative Glomerulonephritis

Glomerular diseases with an MPGN pattern of injury are classified into immune complex (IC-MPGN) and complement-mediated forms (see Chapters 22 and 23).^{50,51} Recurrent MPGN bears major clinical and histologic similarities with the subgroup of patients with CR who have transplant glomerulopathy, and comprehensive assessment of transplant biopsy specimens (see Fig. 113.3 and compare Figs. 112.3 and 112.4), as well as accurate diagnosis of the native kidney disease, is crucial in making this distinction (see Table 113.1). IC-MPGN, rather than transplant glomerulopathy, is suggested by the presence of crescents on light microscopy, stronger staining for C3 and weaker staining for IgM on immunohistologic examination, and subendothelial electron-dense deposits on EM.¹⁸ This distinction has important clinical implications, particularly because recurrence carries a higher risk for subsequent recurrence should retransplantation be considered. A

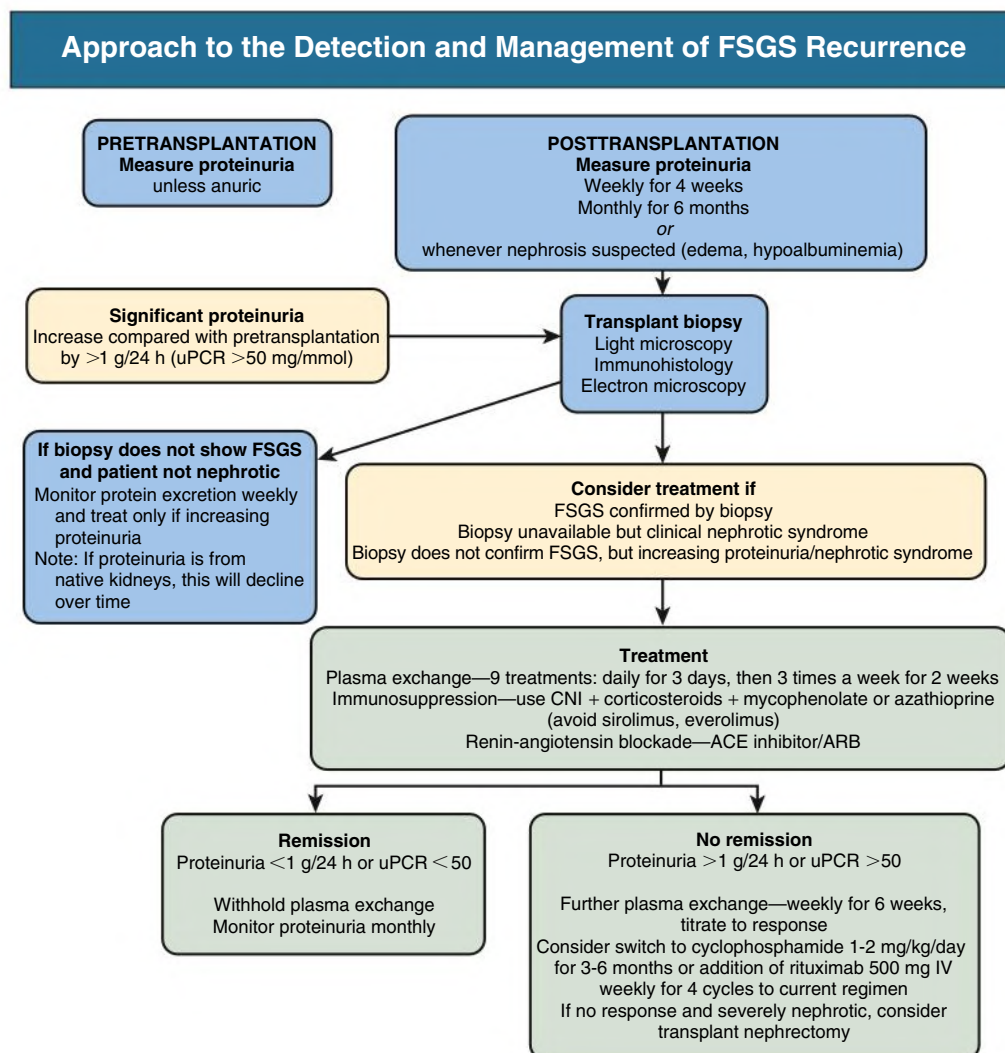


Fig. 113.4 Approach to the Detection and Management of Focal Segmental Glomerulosclerosis (FSGS) Recurrence. Authors' recommendation based on Davenport.⁴⁷ ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CNI, calcineurin inhibitor; uPCR, urine protein-to-creatinine ratio.

retrospective analysis of kidney biopsies that included 70 patients with MPGN suggested that the severity of histologic lesions is predictive of recurrence.⁵²

Immune Complex Membranoproliferative Glomerulonephritis

IC-MPGN appears to be mediated by glomerular deposition of immune complexes, triggered by exposure to endogenous or exogenous (e.g., hepatitis C virus [HCV]) antigens or related to a monoclonal gammopathy. Because the antigens are not necessarily removed by transplantation, recurrence of disease is possible and is seen in 20% to 45% of kidney recipients.^{50,53–56} Graft loss has been reported in up to 50% of those with recurrence, and the risk for recurrence in subsequent grafts approaches 80%.^{50,57–59} Significant geographic diversity in the risk for recurrence is evident, with much higher rates of graft loss resulting from recurrence in areas where the majority of patients with IC-MPGN are HCV positive, such as Spain,⁵³ compared with low-HCV prevalence areas, such as Australia.⁴

No form of treatment for recurrent IC-MPGN has been proven effective, and the underlying cause of MPGN should be considered in each patient. In HCV-associated IC-MPGN, direct-acting antiviral therapy is effective for virus elimination and should provide substantial protection from HCV viremia and IC-MPGN recurrence after transplantation. Other forms of IC-MPGN have been successfully treated with immunosuppression or plasma exchange.

C3 Glomerulopathy

Complement-mediated MPGN, also referred to as C3 glomerulopathy, is a disease process secondary to dysregulated complement activation (see [Chapter 23](#)). There are two subtypes determined by EM appearance: C3 GN and dense deposit disease (DDD).⁵¹ DDD has clinical and histologic similarities to transplant glomerulopathy. It is suggested by granular staining for C3 on capillary loops without immunoglobulin deposits, and the characteristic ribbon-like intramembranous dense deposits on EM (see [Fig. 22.2](#)).^{18,60} DDD has been found to recur in 50% to 80% of grafts, typically manifesting with proteinuria, hematuria, and slowly progressive loss of kidney function.⁶¹ The disease course tends to be slow, and whereas graft loss caused by recurrence is ultimately seen in the majority of patients, this generally occurs beyond the first 10 years after transplantation.^{4,61} Graft loss has been associated with male sex, crescents on biopsy, and heavy proteinuria.⁶¹ No effective therapy is known, and although plasma exchange and immunosuppression have been described, these are not supported by good evidence, and control of blood pressure and proteinuria with RAS blockade is the preferred therapy. Given the role of alternate complement pathway mutations in DDD (see [Chapter 23](#)), administration of complement antagonists such as eculizumab may be beneficial but remain unproven, with inconsistent results reported in published case series to date.^{59,62–64}

Congenital Nephrotic Syndrome

Congenital nephrotic syndrome of the Finnish type has been reported to recur after transplantation and cause graft loss; however, the mechanism of kidney damage is likely different between primary and recurrent disease. The primary disorder in some cases is caused by a mutation of the *NPHS1* gene that results in the complete absence of nephrin, and therefore transplantation causes de novo exposure to nephrin in the transplanted kidney. Neoantigen exposure may cause antibody development and deposition that damages the slit diaphragm and produces a type of MN with podocyte fusion on EM and a clinical picture of heavy proteinuria and ultimately graft failure. Cyclophosphamide-based rescue therapy may be successful.⁶⁵

Antineutrophil Cytoplasmic Antibody–Associated Pauci-Immune Vasculitis

Kidney transplantation for ANCA-associated vasculitis (AAV) is associated with a reduction in the frequency of disease relapse by approximately 50% compared with patients remaining on dialysis, and patient and graft survival posttransplantation is similar to that in other transplant recipients.⁶⁶

Recurrence rates in AAV appear to be declining over time. A 1998 pooled analysis of reported case series examining recurrence of AAV detected recurrence in 17% of the 127 included patients, with kidney involvement demonstrated in approximately 60% and graft losses reported in 25% of these.⁶⁶ More recently, a review of transplant outcomes in AAV show a decline in recurrence rates to 1% to 2% per year in contemporary studies, possibly because of the improvement in immunosuppression regimens since the 1990s.^{67,68}

Clinical parameters have not been found to be useful in predicting those patients most likely to relapse. Pretransplantation disease course, duration of dialysis, ANCA titers at time of transplantation and during follow-up, cytoplasmic ANCA (c-ANCA) or perinuclear ANCA (p-ANCA) pattern, antibody specificity to PR3 or MPO, disease subtype (granulomatous polyangiitis, microscopic polyangiitis, or kidney-limited vasculitis), and donor source have not been found to significantly impact recurrence rates.^{66,67}

The prevention and management of relapse has not been prospectively examined. In most reports, patients did not receive a transplant until they were in clinical remission; however, successful transplantation in the face of persisting ANCA positivity is well recognized. In the absence of firm evidence, we recommend that clinical remission be maintained for 6 to 12 months before transplantation to reduce the risk for recurrence and also to avoid risks associated with performing transplantation in a debilitated patient, especially if kidney failure has occurred soon after presentation with systemic vasculitis.⁶⁹ Monitoring ANCA to indicate AAV recurrence is commonly performed; however, the test performance characteristics in this context have not been reported. Patients with kidney relapses have generally been managed with cyclophosphamide-based regimens, as used for kidney vasculitis in native kidneys, which have been reported to be successful in inducing remission in 11 of 16 (69%) cases.^{66,67} No negative impact of the use of cyclophosphamide to treat relapse has been reported, but cumulative dose, if it had also been used pretransplant, would significantly increase the risk for bladder cancer and infertility. Significant interest exists in the use of rituximab for AAV recurrence posttransplant. Although current literature is limited to case reports, in our opinion, rituximab is a reasonable second-line alternative to cyclophosphamide in this context.^{67,70}

Anti–Glomerular Basement Membrane Disease

Histologic recurrence of anti-GBM disease is seen in 50% of patients who receive a transplant while circulating anti-GBM antibodies persist but rarely when patients undergo transplantation 6 months or more after the disappearance of anti-GBM antibodies. With delayed transplantation, the rate of clinical recurrence is very low, and since the implementation of this practice in Australia, no grafts were lost after transplantation in 47 patients followed for up to 10 years.⁴ Rare episodes of recurrence should be treated as for native kidney disease with corticosteroids, cyclophosphamide, and aggressive plasma exchange (see [Chapter 25](#)).

Recurrent anti-GBM disease is distinct from de novo anti-GBM disease, which is seen in up to 15% of transplant recipients with Alport syndrome who develop anti-GBM antibodies in response to neoantigen exposure (α chain of type IV collagen) via the transplant.⁷¹ This is discussed further in [Chapter 25](#).

Lupus Nephritis

The reported recurrence rate of LN has varied from 2% to 54% depending on the diagnostic criteria used.^{72,73} Recurrence has been reported early (days) and late (years) after transplantation, with a median time to recurrence of approximately 4 years.^{72,73} The clinical and histologic pattern of recurrence is variable but is typically more benign in histology and clinical expression than the patient's original disease. Although the majority of patients with kidney failure caused by lupus have had diffuse proliferative (class III or IV) LN, mesangial proliferative (class II) LN is the most commonly described lesion after transplantation, followed by class III and membranous LN (class V).⁷³ Duration of dialysis and serologic activity before transplantation do not predict recurrence, and antinuclear antibody titer and complement levels are unreliable markers of disease recurrence. There is no consistent relationship between recurrence of nephritis and activity of extrarenal lupus after transplantation.

The long-term outcome for lupus patients after transplantation is controversial but appears similar to that of the general posttransplantation population.^{72,74} Although recurrence is an uncommon cause of graft loss within the first 10 years after transplantation, with no cases of graft loss reported from 86 recipients in one registry analysis,⁴ late graft losses do occur. It is clear that lupus patients, particularly those with a lupus anticoagulant, are at increased risk for thrombotic events after transplantation, including graft thrombosis.⁷⁵

Management of recurrent LN has not been systematically studied; corticosteroids, cyclophosphamide, MMF, and plasma exchange have been used, with variable results reported. Anticoagulation during the perioperative and early posttransplantation phases should be considered for those with a history of thrombosis or lupus anticoagulant positivity. Successful retransplantation has been reported after graft loss caused by recurrence.⁷⁵

Thrombotic Microangiopathy and Hemolytic Uremic Syndrome

Recurrence risk is dependent on the underlying cause of thrombotic microangiopathy (TMA). Typical childhood Shiga toxin–associated HUS seldom recurs, whereas recurrence of atypical HUS (aHUS), which is associated with dysregulation of the alternative complement pathway, occurs in up to 80% of patients depending on the mutations involved.^{76,77} The diagnosis of recurrent TMA, including HUS, is complicated by the fact that *de novo* TMA is seen in 1% to 5% of kidney transplant recipients, most commonly associated with the use of CNIs (both tacrolimus and cyclosporine carry a similar risk), sirolimus, or with acute vascular rejection. Drug-induced *de novo* TMA is generally observed within 14 days of drug commencement.

The risk for aHUS recurrence and its impact on graft survival is dependent on the complement pathway mutation responsible. Mutations in complement factor H are associated with a recurrence risk of approximately 75%, and more than 90% of those with recurrence incur graft failure, typically within the first year.^{76,78} In contrast, mutations of membrane cofactor protein are associated with a recurrence rate of only 20% and substantially better survival rates than in patients with factor H or I mutations.⁷⁹ Given these variations in risk for recurrence, genotypic evaluation of the recipient and any potential living donors before transplantation is advisable for all patients with kidney failure caused by atypical HUS.⁸⁰ Combined liver-kidney transplantation may be considered in selected candidates with defined mutations.⁸¹ Recurrence is generally within the first 6 months after transplantation, thought to be because of ischemia-reperfusion driven endothelial injury; however, late recurrences have been reported.⁷⁷ The clinical presentation may be gradual or abrupt, with thrombocytopenia, hemolysis, and progressive kidney dysfunction.

Historically, treatment for recurrent aHUS has relied on withdrawal of any potential causative agent such as CNIs (with a temporary increase in the dose of corticosteroids to reduce the risk of acute rejection), and plasma exchange. This strategy was only modestly effective, for both first episode and posttransplant aHUS.^{76,82} The development of eculizumab, a humanized monoclonal antibody targeting C5 of the complement membrane attack complex, has revolutionized treatment for aHUS. In the last few years, a number of cohort studies and a large registry analysis have shown that the preemptive use of eculizumab in patients with aHUS leads to greatly reduced recurrence rates and improved graft survival.^{83–85} A meta-analysis of 13 cohort studies and five case series that included a total of 380 patients confirmed protection against recurrence when eculizumab was used preemptively, with recurrence seen in under 6% of cases. Commencement of eculizumab after development of posttransplant aHUS recurrence was also relatively effective, with graft loss reported in approximately 20% of 192 cases.⁸⁶ Both outcomes were vastly superior to historical rates of recurrence and graft loss in those with recurrence. Although the responsible mutations were frequently not identified in this meta-analysis, where it was recorded, the most common mutations were complement factor H and I mutations, which have recurrence rates pre-eculizumab approaching 80%, with graft loss in those with recurrence of more than 90%, typically within the first year.⁷⁸ A subsequent retrospective analysis of the impact of eculizumab found prophylactic use to be superior to initiation after recurrence in terms of graft survival.⁸⁷

We recommend that eculizumab be commenced pretransplant (day of surgery or prior) for high-risk mutations including complement factor H, complement factor B, and C3 and/or previous graft recurrence, and considered for moderate risk mutations such as complement factor I or anticomplement factor H antibody.⁸⁷ This position is supported by the 2016 KDIGO guidelines.⁸⁰

There is less evidence to guide the decision on if and when to cease eculizumab. It is likely that this decision will differ for patients with different gene mutations. Recurrence rates decline after 12 to 24 months,⁷⁶ so we would recommend waiting 24 months posttransplant before attempting cessation. After cessation, intensive monitoring of graft function, hemoglobin and platelet count, blood film, and serum LDH appear warranted in our opinion; however, this strategy has not been scientifically examined. Prompt reintroduction of eculizumab should be considered in cases of recurrence after eculizumab cessation.

Scleroderma

Few patients with scleroderma undergo kidney transplantation. The largest analysis of transplanted scleroderma patients consisted of 86 patients reported to the UNOS registry and showed graft survival of 62% at 1 year, 47% at 5 years after transplantation, and 27% at 10 years; 24% of recipients died during the 10-year observation period.⁸⁸ The recurrence rate could not be accurately determined; however, recurrence was responsible for graft loss in 21% of patients in whom the cause was identified. A more recent analysis of 22 patients from the ANZDATA registry reported similar graft survival of 78% at 1 year, 53% at 5 years, and 28% at 10 years.⁸⁹ The management of scleroderma after transplantation is unstudied; however, the use of RAS blockade after transplantation to treat hypertension would seem appropriate. Overall, kidney transplantation appears to be an appropriate treatment for those with scleroderma and kidney failure, in the absence of contraindications to transplantation such as severe scleroderma gut or lung disease.⁹⁰

AMYLOID, LIGHT-CHAIN DISEASE, AND FIBRILLARY AND IMMUNOTACTOID GLOMERULOPATHIES

Amyloidosis

The risk and impact of recurrence for patients who undergo transplantation because of systemic amyloidosis are dependent on its

cause. Management of AL amyloid is primarily directed at treating the underlying plasma cell dyscrasia, most commonly by high-dose chemotherapy and autologous stem cell or allogeneic bone marrow transplantation. Recurrence after transplantation is likely if the malignancy is not fully controlled but may be susceptible to further chemotherapy. The largest reported series by Heybeli et al. includes 60 patients who received a kidney transplant for AL amyloid-induced kidney failure, roughly half of whom (28 patients) had asymptomatic cardiac involvement.⁹¹ Before kidney transplant, 51 of the 60 patients had received treatment for amyloidosis, which resulted in complete remission for 37 (62%) and partial remission in 11 (19%) patients.⁹¹ Of 9 patients who were treatment naïve at the time of kidney transplant, 7 were treated with melphalan and autologous stem cell transplant after kidney transplant, all of whom had complete remission.⁹¹ Death-censored graft survival was 98.3% at 1 year and 95.8% at 5 years. No grafts were lost through recurrence.⁹¹ Protocol biopsies were performed out to 10 years and revealed amyloid recurrence in the graft in 13 (22%) patients, with an estimated median time to recurrence of 122 months.⁹¹ Recurrence was predominantly observed in those with a poor response to initial chemotherapy. This paper built on previous work with small patient cohorts that revealed similar findings.^{92,93} Thus transplantation appears to be a viable option for patients with kidney failure caused by AL amyloidosis, provided they have achieved a good hematologic response to treatment.⁹⁰

Secondary (AA) amyloidosis is typically a more insidious disease; kidney failure is a relatively frequent complication, and kidney transplantation is frequently effective.⁹² The risk for recurrence of AA amyloid depends on the ability to eradicate the underlying cause of chronic inflammation. In AA amyloid, recurrence as a result of chronic infection is unlikely if the infection can be eradicated pretransplantation, whereas conditions such as rheumatoid arthritis may persist after transplantation and lead to recurrent amyloid in the graft. Less than 5% of patients with AA amyloid associated with familial Mediterranean fever (which can be managed with colchicine after transplantation) have recurrence at 10 years after transplantation.⁹⁴ A Norwegian series of 62 transplants in patients with AA amyloid, mostly secondary to rheumatic diseases, demonstrated a recurrence rate of 10% at an average of 5 years of follow-up, with recurrence causing graft loss in two patients only. Overall, patient and graft survival rates were 65% and 62%, respectively, at 5 years, with most losses resulting from infection.⁹⁵

Light-Chain Nephropathy

Patients with light-chain nephropathy have occasionally received transplants, and recurrence is common; one case series reported recurrence in five of seven patients at a range of 2 to 45 months after transplantation. Those with recurrence developed proteinuria, hypertension, and progressive graft dysfunction. One of seven has had long-term graft function, and one died soon after transplantation because of myeloma.⁹⁶ Kidney transplantation is therefore generally inadvisable for patients with this disease, unless performed in conjunction with bone marrow transplantation as a means of curing the underlying disease. Rituximab and bortezomib have been used with success in patients with light-chain nephropathy recurrence in the graft, but insufficient evidence exists to recommend either agent as a standard of care.

Fibrillary and Immunotactoid Glomerulopathies

Fibrillary glomerulopathy (DNAJB9 Deposition Disease) and immunotactoid glomerulopathies are known to recur in approximately 50% of those undergoing transplantation for these diseases, and although early graft loss as a result of recurrence has been reported, decline in

graft function is most commonly slow and does not appear to have an impact on 5-year graft survival rates.⁹⁷

RECURRENCE OF METABOLIC DISEASES AFFECTING THE KIDNEY TRANSPLANT

Diabetes Mellitus

Diabetes mellitus is the most common cause of kidney failure in most parts of the world. However, these patients undergo transplantation less commonly than those with GN because of a higher prevalence and severity of CV comorbidity and inferior survival in registry studies.^{98,99} Recurrence of diabetic nephropathy affects at least 25% of recipients at an average follow-up of 6 years and with some patients diagnosed within 3 years of transplantation.¹⁰⁰ Histologic and clinical features are similar to those of native kidney diabetic nephropathy. The risk of graft loss from recurrence has not been well documented but appears to be significantly less than with GN, probably because of the competing risk of death from CV disease. NODAT is also common and has been reported to cause nephropathy in the graft, also manifesting within 5 years of transplantation in a small proportion of cases. The extent to which this contributes to graft loss also awaits clarification, although one registry analysis suggested that the impact on graft failure is insignificant compared with the impact on premature death with a functioning graft.⁵

Primary Hyperoxaluria

Primary hyperoxaluria (see [Chapter 60](#)) is a rare autosomal recessive disease caused by defective or absent hepatic production of alanine glyoxylate aminotransferase, resulting in systemic accumulation of calcium oxalate (oxalosis). Kidneys and blood vessels in particular are affected. Kidney transplantation alone is frequently complicated by hyperoxaluria and rapid deposition of oxalate crystals within the graft causing graft failure within weeks to months from transplantation ([Fig. 113.5](#)). By contrast, combined liver-kidney transplantation corrects the underlying metabolic deficit and permits long-term kidney graft survival, provided the total body burden of oxalate present at the time of transplantation can be managed. In an analysis of the USRDS that included 190 adults with oxalosis who went on to undergo kidney transplantation, 134 patients who received a kidney transplant alone experienced 48% 8-year death-censored graft survival. This was inferior to 56 patients who also received a liver (76%) and inferior to a control group with a primary diagnosis of GN (61%).¹⁰¹ Aggressive removal of residual oxalate before transplantation by dialysis and after transplantation by maintenance of high urine volumes, urinary alkalization,

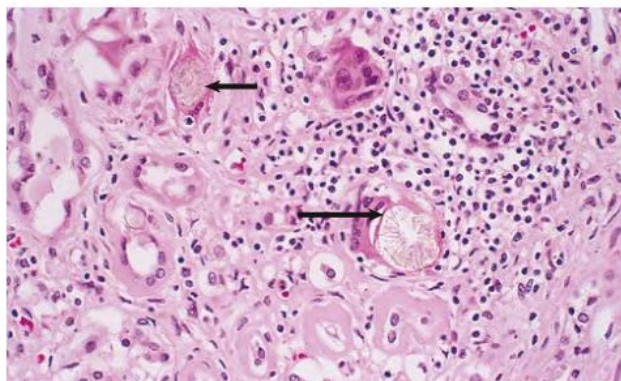


Fig. 113.5 Recurrent Primary Hyperoxaluria. Light microscopy demonstrates oxalate crystals within the tubular lumen (arrows) with a secondary interstitial inflammatory infiltrate.

and pyridoxine supplementation for pyridoxine-sensitive patients also may decrease the risk for graft damage.

Fabry Disease

Fabry disease results from a defect in the lysosomal α -galactosidase A enzyme, which results in tissue accumulation of trihexosylceramide, eventually causing kidney failure (see [Chapter 48](#)). Recurrent Fabry disease has been documented within the graft; however, graft survival does not appear to be affected. Treatment with recombinant α -galactosidase A, should this become widely available, is likely to reduce the risk for kidney failure and also for recurrence.

RECURRENCE OF VIRUS-ASSOCIATED NEPHROPATHIES AND TUMORS IN THE TRANSPLANTED KIDNEY

Virus-associated kidney diseases may recur after transplantation. Hepatitis B and C virus-associated IC-MPGN and MN are known to recur; however, the risk for this can be substantially decreased by successful antiviral therapy before transplantation (see [Chapter 58](#)).

Retransplantation has been reported in patients experiencing graft loss because of BK virus nephropathy. In the largest series with documented recurrence, BK viral replication was found in 11 of 31 (35%) recipients, and BK virus-associated nephritis (BKVAN) in 2

of the 31 (6%) at an median follow-up of 30 months after retransplantation.¹⁰² Rates of BK recurrence were lower when viral clearance had been achieved before retransplantation, with recurrence seen in 15% of BK-negative recipients versus 100% who remained BK positive at the time of transplantation.¹⁰² In a recently published study of the UNOS/OPTN database between 2005 and 2016, there were 46 (14%) regraft failures in 341 patients who had lost their first graft because of BKVAN. Of these, 7 (15%) lost their regraft because of BKVAN compared with 55 (2%) of the 2458 regrafts who had lost a previous graft for another reason.¹⁰³ There was no difference in overall regraft survival between those who lost their first graft to BKVAN compared with another cause over the median follow-up time of 4.7 years.¹⁰³ Thus, retransplantation, ideally delayed until BK virus is not detectable in serum by polymerase chain reaction, appears to be safe and effective.

Patients with posttransplantation lymphoproliferative disease (PTLD) who incur graft loss secondary to direct infiltration or rejection after the withdrawal of immunosuppression may safely and successfully undergo retransplantation after they recover. In an analysis of patients with second kidney transplants in the UNOS registry between 2000 and 2019, 254 patients were found to have had PTLD in their first transplant. These patients were found to have higher rates of PTLD in their second transplant (2.8% vs. 0.1% without previous PTLD, $P = .001$), but there was no difference in graft or patient survival at a median of 4.5 years of follow-up.¹⁰⁴

SELF-ASSESSMENT QUESTIONS

- Risk for graft loss because of recurrent disease is highest for kidney transplant recipients whose kidney failure was caused by:
 - IgAN.
 - FSGS.
 - Type 1 diabetes.
 - Lupus nephritis.
 - Reflux nephropathy.
- A diagnosis of recurrent disease affecting the allograft requires which of the following pieces of information?
 - Allograft biopsy confirming the disease
 - Biopsy diagnosis of the native kidney disease
 - Clinical features of the original kidney disease recurring after transplant
 - Allograft failure caused by the disease
 - Both A and B
- Which one of the following statements is *true* regarding management of recurrent FSGS?
 - First-line therapy includes a 3-day course of daily intravenous pulses of methylprednisolone.
 - Cyclophosphamide should be started once recurrence is confirmed.
 - In a patient who develops nephrotic-range proteinuria 2 weeks after transplant, a biopsy specimen that is normal on light microscopy excludes the diagnosis of recurrent FSGS.
 - Plasma exchange should be started once a diagnosis of recurrent FSGS is made, with a plan to complete approximately nine cycles of therapy.

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